



Course code: MR25-2PG0107

Course Name: Deep Learning & Its Applications

Specialisation : **ML & DL**

## QUESTION BANK

| Q. no         | Questions                                                                                                                                                                  |
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| <b>UNIT-1</b> |                                                                                                                                                                            |
| 1             | Explain the role of sigmoid neurons in feed forward neural networks.                                                                                                       |
| 2             | What is a perceptron, and how does the perceptron learning algorithm work?                                                                                                 |
| 3             | What are multilayer perceptrons (MLPs), and how do they differ from single-layer perceptrons?                                                                              |
| 4             | How do feed forward neural networks achieve their representation power?                                                                                                    |
| 5             | What is back propagation, and how is it used to train feed forward neural networks?                                                                                        |
| 6             | Write the differences between Feed Forward and Feed Backward Neural networks.                                                                                              |
| 7             | What is Deep learning? Explain its uses and applications.                                                                                                                  |
| 8             | Discuss about single layer perceptrons in detail.                                                                                                                          |
| 9             | Differentiate between machine learning and deep learning in detail with examples.                                                                                          |
| 10            | How does the McCulloch-Pitts neuron model capture the basic functionality of a biological neuron, and what are the key components and processes involved in its operation? |
| <b>UNIT-2</b> |                                                                                                                                                                            |
| 11            | What is the difference between Stochastic Gradient Descent and Batch Gradient Descent?                                                                                     |
| 12            | How does Momentum Based Gradient Descent work and how is it different from other optimization algorithms?                                                                  |
| 13            | How does basis transformation using principal component analysis (PCA) help in reducing the dimensionality of data in deep learning?                                       |
| 14            | Explain the Adam optimisation algorithm in detail.                                                                                                                         |
| 15            | How do the various optimization algorithms (GD, momentum-based GD, Nesterov accelerated GD, AdaGrad, RMSProp, Adam) impact the convergence rate?                           |
| 16            | Write about RMS optimization algorithm in detail.                                                                                                                          |
| 17            | Explain the concept of singular value decomposition (SVD) and how it is used in deep learning.                                                                             |
| 18            | How are eigenvalues and eigenvectors utilized in deep learning, and what                                                                                                   |

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|    | significance do they hold in the analysis of matrices, particularly in tasks such as principal component analysis and understanding the behaviour of neural network layers?                                                                                                                                                                                                                |
| 19 | How does eigenvalue decomposition contribute to the analysis of gradient descent in deep learning?                                                                                                                                                                                                                                                                                         |
| 20 | Explain in detail about the concept of gradient based learning.                                                                                                                                                                                                                                                                                                                            |
|    | <b>UNIT-3</b>                                                                                                                                                                                                                                                                                                                                                                              |
| 21 | What are autoencoders, and how do they relate to principal component analysis (PCA) in terms of dimensionality reduction?                                                                                                                                                                                                                                                                  |
| 22 | How do sparse autoencoders differ from traditional autoencoders, and what are some of the applications of sparse autoencoders in deep learning?                                                                                                                                                                                                                                            |
| 23 | How does bias-variance trade-off can be managed using regularization techniques such as L2 regularization?                                                                                                                                                                                                                                                                                 |
| 24 | How do the practices of introducing noise at the input and implementing the dropout regularization technique contribute to mitigating overfitting and improving the generalization performance of a model? Explore the trade-offs, advantages, and limitations associated with each approach, and offer instances where these methods have proven to be effective in various applications. |
| 25 | Explore the concept of dataset augmentation in deep learning and how it contributes to improving model generalization.                                                                                                                                                                                                                                                                     |
| 26 | What is the role of denoising autoencoders in deep learning, and how do they enhance the robustness of a model?                                                                                                                                                                                                                                                                            |
| 27 | Explain the concept of contractive autoencoders and how they differ from traditional autoencoders.                                                                                                                                                                                                                                                                                         |
| 28 | Describe the concepts of parameter sharing and tying in the context of autoencoders.                                                                                                                                                                                                                                                                                                       |
| 29 | Discuss the concept of early stopping as a regularization technique in deep learning.                                                                                                                                                                                                                                                                                                      |
| 30 | Compare and contrast the effects of injecting noise at the input layer and using dropout regularization in deep learning models.                                                                                                                                                                                                                                                           |
|    | <b>UNIT-4</b>                                                                                                                                                                                                                                                                                                                                                                              |
| 31 | What are some of the key advantages and limitations of popular activation functions and how can we develop better activation functions that are more effective for specific applications?                                                                                                                                                                                                  |
| 32 | What are Learning Vectorial Representations of Words, and how can techniques such as Word2Vec be used to generate high-quality word embeddings that capture the semantic relationships between different words in a corpus?                                                                                                                                                                |
| 33 | Explain the concept of greedy layer wise pre-training, and how it is used to improve the training of deep neural networks?                                                                                                                                                                                                                                                                 |
| 34 | How can we use advanced weight initialization methods to improve the performance of Deep Learning models?                                                                                                                                                                                                                                                                                  |

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| 35            | How does Batch Normalization work, and how can we use this technique to improve the training stability and generalization performance of Deep Learning models?                                                                                                                                                                                    |
| 36            | Discuss scenarios where Greedy Layerwise Pre-training is particularly advantageous and its impact on model convergence.                                                                                                                                                                                                                           |
| 37            | Discuss the significance of using advanced activation functions in neural networks. How do alternatives like ReLU and variants address the limitations of traditional functions?                                                                                                                                                                  |
| 38            | How do techniques like He initialization and Xavier/Glorot initialization contribute to faster convergence and improved model performance? Discuss scenarios where choosing an appropriate weight initialization method is crucial.                                                                                                               |
| 39            | Describe the process by which the Batch Normalization technique normalizes activations within intermediate layers during training. Explore the advantages it provides in terms of hastening convergence and enhancing the generalization performance of a model. Discuss practical scenarios where Batch Normalization is particularly effective. |
| 40            | How does the Word2Vec algorithm contribute to learning vector representations of words? Explain the underlying principles of continuous bag-of-words (CBOW) and skip-gram models within Word2Vec. Discuss the practical applications and benefits of utilizing Word2Vec.                                                                          |
| <b>UNIT-5</b> |                                                                                                                                                                                                                                                                                                                                                   |
| 41            | How can Convolutional Neural Networks (CNNs) be used to extract high-level features from image and video data, and what are some of the key technical challenges involved in this process?                                                                                                                                                        |
| 42            | What are Gated Recurrent Units (GRUs) and Long Short-Term Memory (LSTM) cells, and how can these specialized RNN architectures can be used to model long-term dependencies and avoid the vanishing gradient problem?                                                                                                                              |
| 43            | Explore the GoogLeNet architecture and elucidate some of its significant contributions to the field of deep learning.                                                                                                                                                                                                                             |
| 44            | How can Long Short term Memory (LSTM) be used to extract high-level features from image and video data, and what are some of the key technical challenges involved in this process?                                                                                                                                                               |
| 45            | Examine the architecture of AlexNet in the realm of deep learning. Highlight key aspects that have significantly impacted the field, showcasing the model's advancements and innovations.                                                                                                                                                         |
| 46            | How can Recurrent Neural Networks (RNNs) and Back propagation through time (BPTT) be used to model temporal dependencies in sequential data such as speech, text and music?                                                                                                                                                                       |
| 47            | Examine the ResNet architecture and highlight key contributions it has made to the field of deep learning.                                                                                                                                                                                                                                        |
| 48            | Explore the applications of Deep Dream and Deep Art in the field of deep learning. How do these techniques leverage neural networks to generate visually captivating and artistic images?                                                                                                                                                         |

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| 49 | How does Guided Backpropagation enhance interpretability in deep neural networks? Discuss the key principles behind this technique and provide examples of scenarios where it is beneficial for visualizing and understanding feature importance in deep learning models.                   |
| 50 | Explain the concept of attention mechanisms in neural networks. How do attention mechanisms enhance model performance by selectively focusing on specific parts of input sequences? Discuss applications and use cases where attention mechanisms have proven to be particularly effective. |



# MALLA REDDY UNIVERSITY

**Subject Name: Generative AI & Its Applications    Subject Code: MR25-2CB0108**

**Year & Semester: I Year M. Tech – II Semester**

## **Unit-Wise Question Bank**

| <b>Qno</b>    | <b>Question</b>                                                                                                                                                                                                            |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UNIT-1</b> |                                                                                                                                                                                                                            |
| 1             | Define Generative AI and critically differentiate discriminative versus generative modeling paradigms using their probabilistic formulations ( $P(Y X)$ vs. $P(X,Y)$ ), supported by suitable examples.                    |
| 2             | Trace the evolution of Generative AI from statistical n-gram models to modern foundation models. Critically analyze the architectural and training-paradigm shifts that enabled GPT, DALL-E, Stable Diffusion, and Gemini. |
| 3             | Analyze the deployment of Generative AI across healthcare, finance, and education. For each domain, evaluate the benefits, risks, and regulatory considerations involved.                                                  |
| 4             | Critically evaluate the role of foundation models and transfer learning in reducing the cost of building task-specific AI systems, supporting your answer with concrete examples.                                          |
| 5             | Compare rule-based, statistical, and deep-learning-based generative approaches. Justify, with reasoning, which approach is best suited for open-domain text generation.                                                    |
| 6             | Explain, with a probabilistic formulation, how a generative model learns to approximate the underlying data distribution. Discuss the consequences of a poor distributional approximation.                                 |
| 7             | Critically analyze the major challenges of Generative AI - hallucination, bias, and computational cost - and propose specific engineering mitigation strategies for each.                                                  |
| 8             | Construct a taxonomy of Generative AI models (Autoregressive, VAE, GAN, Diffusion, Flow-based). For each category, evaluate its strengths, weaknesses, and most suitable application scenario.                             |
| 9             | Critically discuss the historical evolution of Generative AI, evaluating the paradigm shift from Markov Chains and RNNs to attention-based Transformer architectures.                                                      |
| 10            | Evaluate the ethical and societal implications of Generative AI, including deepfakes, misinformation, and job displacement. Propose a framework for its responsible deployment.                                            |
| <b>UNIT-2</b> |                                                                                                                                                                                                                            |
| 11            | Derive the Evidence Lower Bound (ELBO) for a VAE from first principles. Critically analyze the trade-off between the reconstruction term and the KL-divergence regularization term.                                        |
| 12            | Critically analyze the GAN minimax objective. Derive the optimal discriminator for a fixed generator and show that, at equilibrium, the generator's distribution matches the true data distribution.                       |
| 13            | Evaluate DCGAN, CycleGAN, StyleGAN, and Conditional GAN architectures. For each, critically assess its architectural innovation, the specific limitation it addresses, and its suitability for a given application.        |
| 14            | Derive the reparameterization trick mathematically and critically justify why it is essential for end-to-end gradient-based training of VAEs.                                                                              |
| 15            | Critically analyze the mode collapse problem in GANs. Evaluate at least three mitigation techniques (e.g., minibatch discrimination, WGAN, unrolled GANs) and compare their effectiveness.                                 |
| 16            | Evaluate the Frechet Inception Distance (FID) and Inception Score as generative-model evaluation metrics. Critically discuss their limitations and propose complementary evaluation strategies.                            |
| 17            | Derive the Wasserstein distance formulation used in WGAN. Critically analyze how it addresses vanishing gradients and training instability relative to a vanilla GAN.                                                      |
| 18            | Critically compare VAEs and GANs across sample quality, training stability, latent-space interpretability, and mode coverage. Justify which is preferable for a specified use case (e.g.,                                  |

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|    | anomaly detection vs. photorealistic image synthesis).                                                                                                                                                           |
| 19 | Analyze the vanishing-gradient and mode-collapse failure modes encountered in adversarial training. Propose and justify architectural or training modifications that mitigate them.                              |
| 20 | Critically evaluate Conditional VAEs (CVAE) for controlled generation. Design a CVAE-based solution for a specified real-world scenario (e.g., attribute-controlled face generation).                            |
|    | <b>UNIT-3</b>                                                                                                                                                                                                    |
| 21 | Derive the Scaled Dot-Product Attention formula from first principles. Critically justify the $1/\sqrt{d_k}$ scaling factor and analyze what happens to training if it is omitted.                               |
| 22 | Critically evaluate the MLM, CLM, and Seq2Seq pre-training paradigms for LLMs. Justify, with supporting evidence, why CLM is best suited for open-ended text generation.                                         |
| 23 | Analyze the role of layer normalization and residual connections in stabilizing deep Transformer training. Critically evaluate Pre-LN versus Post-LN configurations.                                             |
| 24 | Critically evaluate the design choices in the Transformer feed-forward sublayer, including the role of GELU/SwiGLU activations, and their impact on model expressivity.                                          |
| 25 | Compare absolute, relative, and rotary (RoPE) positional encoding schemes. Critically evaluate why RoPE is favored in modern LLMs for long-context generalization.                                               |
| 26 | Critically analyze causal and padding attention masks with worked examples, evaluating their role in enabling correct, parallelized Transformer training.                                                        |
| 27 | Critically evaluate the Kaplan and Chinchilla scaling laws for LLMs. Given a fixed compute budget, justify how model size and dataset size should be balanced.                                                   |
| 28 | Critically evaluate Instruction Tuning and Reinforcement Learning from Human Feedback (RLHF) as alignment techniques. Analyze their respective strengths, limitations, and failure modes (e.g., reward hacking). |
| 29 | Critically compare encoder-only, decoder-only, and encoder-decoder Transformer architectures. Justify the architectural choice for three distinct NLP tasks.                                                     |
| 30 | Critically evaluate techniques for extending the Transformer context window (sliding-window attention, sparse attention, RoPE interpolation), analyzing the trade-offs in compute and accuracy.                  |
|    | <b>UNIT-4</b>                                                                                                                                                                                                    |
| 31 | Critically evaluate zero-shot, one-shot, and few-shot prompting strategies. Design an experiment to determine which strategy is most effective for a specified classification task.                              |
| 32 | Critically analyze Chain-of-Thought (CoT) prompting. Evaluate why it improves multi-step reasoning performance and identify scenarios where it fails.                                                            |
| 33 | Critically compare full fine-tuning, LoRA, Adapters, and prompt tuning for adapting a pre-trained LLM. Justify the best approach under a limited compute and data budget.                                        |
| 34 | Critically evaluate the Retrieval-Augmented Generation (RAG) architecture. Analyze how retrieval quality affects generation quality and propose strategies to improve retrieval precision.                       |
| 35 | Critically evaluate the role of embeddings and vector databases in a RAG pipeline. Compare at least two similarity-search algorithms (e.g., HNSW, IVF) for large-scale retrieval.                                |
| 36 | Critically analyze In-Context Learning as an emergent capability of LLMs. Evaluate competing hypotheses that attempt to explain why it works.                                                                    |
| 37 | Critically evaluate agent-based orchestration frameworks employing tool use and function calling. Design a multi-step agentic workflow for a specified real-world task.                                          |
| 38 | Critically analyze prompt injection attacks on LLM-based applications. Evaluate at least three mitigation strategies and discuss their trade-offs.                                                               |
| 39 | Critically evaluate hallucination-mitigation techniques (grounding, RAG, fact verification, self-consistency). Justify which combination is most robust for a high-stakes application.                           |
| 40 | Critically evaluate BLEU, ROUGE, and perplexity as LLM evaluation metrics. Justify why human evaluation remains necessary and propose a hybrid evaluation framework.                                             |
|    | <b>UNIT-5</b>                                                                                                                                                                                                    |
| 41 | Derive the forward and reverse diffusion processes mathematically. Critically analyze the role of the noise schedule in determining generation quality.                                                          |
| 42 | Critically compare DDPM and DDIM sampling strategies, evaluating the trade-off between sampling speed and sample quality.                                                                                        |

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| 43 | Critically evaluate the Stable Diffusion architecture (U-Net, VAE, CLIP text encoder), analyzing how cross-attention conditions image generation on text prompts.                                                        |
| 44 | Critically evaluate multimodal generative models such as CLIP, DALL-E, and GPT-4V, analyzing how contrastive learning achieves text-image alignment.                                                                     |
| 45 | Critically evaluate Latent Diffusion Models, analyzing the computational trade-offs of operating in latent space versus pixel space.                                                                                     |
| 46 | Critically evaluate responsible AI practices - bias mitigation, watermarking, and content provenance standards such as C2PA - for generative systems. Propose a governance framework for a generative-AI product launch. |
| 47 | Critically evaluate deepfake detection techniques, analyzing the ongoing arms race between generation and detection methods.                                                                                             |
| 48 | Critically evaluate model compression techniques (quantization, knowledge distillation, pruning) for deploying large generative models. Justify the trade-offs between latency and output quality.                       |
| 49 | Critically evaluate the copyright and intellectual-property challenges associated with training data used in Generative AI. Analyze current legal approaches and propose a fair-compensation framework.                  |
| 50 | Critically evaluate emerging trends in Generative AI - agentic AI, multimodal foundation models, and AI regulation. Predict the trajectory of the field over the next five years, justifying your reasoning.             |



**Subject Name: COMPUTER VISION AND ITS APPLICATIONS**

**Subject Code: MR25-2CB0046**

**Unit-Wise Question Bank**

| <b>Q.No</b>   | <b>Question</b>                                                                                                                                           |
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| <b>UNIT-1</b> |                                                                                                                                                           |
| 1             | Define computer vision and explain the concept of vision as recognition.                                                                                  |
| 2             | Given a $5 \times 5$ grayscale image and a $3 \times 3$ kernel, compute the convolution output and interpret the effect of the kernel on image smoothing. |
| 3             | Analyze the difference between sequential and parallel image processing operations.                                                                       |
| 4             | An image contains multiple overlapping objects. Describe how object location techniques can be used to identify each object.                              |
| 5             | Perform grayscale to binary image conversion using an appropriate threshold and explain the obtained output.                                              |
| 6             | Compare grayscale and color images in terms of storage, processing, and applications.                                                                     |
| 7             | Give advantages and disadvantages of sequential and parallel image processing operations.                                                                 |
| 8             | Evaluate the role of convolution and Point Spread Function (PSF) in image restoration.                                                                    |
| 9             | Design a simple image preprocessing pipeline using grayscale conversion, filtering, and binary image operations for an industrial inspection system.      |
| 10            | Explain Convolutions and Point spread Functions with example                                                                                              |
| <b>UNIT-2</b> |                                                                                                                                                           |
| 11            | Perform a 2D affine transformation on given coordinates and compute the transformed output.                                                               |
| 12            | Analyze the role of intrinsic and extrinsic camera parameters in geometric transformations.                                                               |
| 13            | Develop a comprehensive model integrating optics, lighting, and sensor characteristics for realistic image synthesis.                                     |
| 14            | Define image formation and explain its significance in computer vision.                                                                                   |
| 15            | List the types of geometric transformations used in computer vision.                                                                                      |
| 16            | Describe the stages involved in digital image formation.                                                                                                  |
| 17            | Using MATLAB/OpenCV, implement image rotation and scaling and compare the outputs.                                                                        |
| 18            | Compare perspective projection and orthographic projection with suitable diagrams.                                                                        |

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| 19 | Evaluate the impact of sampling and aliasing on digital image quality.                                                         |
| 20 | Write a short notes on “3D transformations and 3D rotations 3D to 2D projections”                                              |
|    | <b>UNIT-3</b>                                                                                                                  |
| 21 | Define Gaussian filter and Median filter.                                                                                      |
| 22 | Analyze the role of morphological operations in shape preservation and noise elimination.                                      |
| 23 | Explain the working principle of Gaussian smoothing.                                                                           |
| 24 | Describe mathematical morphology and its applications.                                                                         |
| 25 | Apply Median filtering to remove salt-and-pepper noise from an image.                                                          |
| 26 | Implement dilation and erosion using OpenCV/MATLAB and compare the results.                                                    |
| 27 | Compare Gaussian, Median, and Rank Order filters for noise removal.                                                            |
| 28 | Analyze the role of morphological operations in industrial vision applications.                                                |
| 29 | Evaluate the effectiveness of sharpening techniques in enhancing image quality.                                                |
| 30 | Design an image enhancement algorithm using filtering and morphological operations to detect defects in manufactured products. |
|    | <b>UNIT-4</b>                                                                                                                  |
| 31 | Define thresholding and image segmentation.                                                                                    |
| 32 | List different threshold selection methods.                                                                                    |
| 33 | Explain adaptive thresholding with an example.                                                                                 |
| 34 | Describe region-growing methods used in image segmentation.                                                                    |
| 35 | Apply Otsu's thresholding method to segment an image.                                                                          |
| 36 | Implement adaptive thresholding using MATLAB/OpenCV and compare it with global thresholding.                                   |
| 37 | Compare global and local thresholding methods.                                                                                 |
| 38 | Analyze the effect of illumination changes on threshold selection.                                                             |
| 39 | Evaluate variance-based and entropy-based thresholding techniques.                                                             |
| 40 | Develop a segmentation algorithm for automatic detection of defects in industrial components using adaptive thresholding.      |
|    | <b>UNIT-5</b>                                                                                                                  |
| 41 | Define edge detection and list different edge detection operators.                                                             |
| 42 | Explain the theory of 3×3 template operators with examples.                                                                    |
| 43 | State the principle of the Canny edge detector.                                                                                |
| 44 | Apply the Sobel operator to detect horizontal and vertical edges in an image.                                                  |

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| 45 | Implement Canny edge detection using OpenCV/MATLAB and compare it with the Laplacian operator.                                    |
| 46 | Compare Sobel, Prewitt, Roberts, and Canny edge detection methods.                                                                |
| 47 | Analyze the influence of threshold values on edge detection performance.                                                          |
| 48 | Evaluate the performance of differential gradient operators in noisy images.                                                      |
| 49 | Design a complete edge detection system for automatic object boundary extraction in quality inspection using the Canny algorithm. |
| 50 | Describe hysteresis thresholding in the Canny edge detector.                                                                      |



# MALLA REDDY UNIVERSITY

**Subject Name: Big Data Analytics**

**Subject Code: MR25-2CB0047**

**Year & Semester: I Year M. Tech – II Semester**

## **Unit-Wise Question Bank**

| <b>Q.No</b>   | <b>Question</b>                                                                                                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UNIT-1</b> |                                                                                                                                                                                                                                                |
| 1             | a) Define Data Analytics and give its importance.<br>b) Describe the key steps involved in the Process of Data Analytics.                                                                                                                      |
| 2             | Discuss the four main types of data analytics: descriptive, diagnostic, predictive, and prescriptive. Include a real-world example for each.                                                                                                   |
| 3             | What are the four levels of data measurement? Describe each with examples                                                                                                                                                                      |
| 4             | How does data visualization facilitate better decision-making? Give examples of common visualization techniques.                                                                                                                               |
| 5             | Define and calculate any four measures of central tendency using a sample dataset.                                                                                                                                                             |
| 6             | Apply different measures of dispersion to a sample dataset, and analyze the results with examples.                                                                                                                                             |
| 7             | Explain Skewness and Kurtosis. Differentiate between positive and negative skewness with examples                                                                                                                                              |
| 8             | How graphical techniques such as box plots help summarize data distributions? Discuss how to interpret skewness and kurtosis.                                                                                                                  |
| 9             | Discuss the Central Limit Theorem. Explain its significance and provide real-world examples of its application.                                                                                                                                |
| 10            | What is a confidence interval, and how is it constructed to estimate a population parameter?                                                                                                                                                   |
| <b>UNIT-2</b> |                                                                                                                                                                                                                                                |
| 11            | Analyze the different types of probability distributions, apply them to practical examples, and discuss their characteristics.                                                                                                                 |
| 12            | a)What do you understand by the term "sampling distribution"? State Factors Influencing Sampling Distribution.<br>b)Differentiate between the sampling distribution of the mean and the sampling distribution of the proportion with examples. |
| 13            | a)Discuss Hypothesis testing with suitable example.<br>b)Explain in detail developing of null and alternative hypothesis.                                                                                                                      |
| 14            | Describe the procedure of hypothesis testing with a real-life example (e.g., testing average marks of students).                                                                                                                               |
| 15            | What is a scatter diagram? Explain how it is constructed and interpreted. Illustrate your answer with a suitable example.                                                                                                                      |
| 16            | Discuss the process of Two-Way ANOVA in testing the effects of two factors. Provide a standard ANOVA summary table and evaluate how the F-ratio helps in drawing conclusions.                                                                  |

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| 17 | Define the Chi-Square test and discuss its purpose in statistical hypothesis testing. Explain its types.                                                           |
| 18 | Illustrate the purpose of data imputation in data cleaning. Describe any four common data imputation techniques with examples.                                     |
| 19 | Discuss the importance of data cleaning in statistical analysis. Describe common data cleaning steps including handling missing values and outlier detection.      |
| 20 | Using a sample dataset, show how correlation and regression analysis can be used to understand relationships between variables.                                    |
|    | <b>UNIT-3</b>                                                                                                                                                      |
| 21 | Explain the use of Text to Columns and Concatenate functions in Excel with practical examples.                                                                     |
| 22 | Differentiate between absolute and relative cell references and their impact on formula replication.                                                               |
| 23 | Demonstrate the use of Excel functions to concatenate text strings with the RIGHT function.                                                                        |
| 24 | How would you apply data validation in Excel to maintain data integrity? Provide examples                                                                          |
| 25 | Describe important time and date calculation techniques in Excel.                                                                                                  |
| 26 | How can conditional formatting be used effectively to highlight key insights in a dataset? Illustrate with suitable example.                                       |
| 27 | Explain how you would clear and reset formatting styles in Excel without deleting data.                                                                            |
| 28 | How can conditional formatting be used to hide cells in a worksheet? Explain with a scenario.                                                                      |
| 29 | Discuss the concept of Excel Styles, illustrating your answer with an example, and explain how they contribute to effective and consistent spreadsheet formatting. |
| 30 | Demonstrate with examples how to create, apply, and modify cell styles to enhance the readability and consistency of large data sets.                              |
|    | <b>UNIT-4</b>                                                                                                                                                      |
| 31 | Define a Pivot Table and explain its purpose in Excel with suitable examples                                                                                       |
| 32 | List the steps involved in creating a PivotTable using a given dataset. Illustrate with an example.                                                                |
| 33 | How to specify data sources for a PivotTable? What are the options available for selecting data?                                                                   |
| 34 | Explain how pivot charts complement pivot tables with visual representation of data                                                                                |
| 35 | Apply filtering and sorting techniques in a PivotTable to display top 5 products based on total sales. Describe each step.                                         |
| 36 | Describe the use of grouping items in pivot tables and explain a practical use case.                                                                               |
| 37 | Analyze the steps to update and refresh PivotTable data when the source data changes. Why is this process important?                                               |
| 38 | Evaluate how using Slicers improves interactivity and data visualization in PivotTables and PivotCharts. Provide examples.                                         |
| 39 | Compare and contrast different chart types (Column, Line, Pie, and Bar). Discuss scenarios where each chart type is most effective.                                |
| 40 | Describe how to modify charts (change color, labels, titles, and chart elements) to make them more informative and visually appealing.                             |
|    | <b>UNIT-5</b>                                                                                                                                                      |
| 41 | Define an Oracle Database and explain its key features and architecture components                                                                                 |

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| 42 | a) Describe the structure and purpose of the SQL SELECT statement with examples<br>b) Write a simple SELECT statement to retrieve employee names and salaries. Modify your query to sort results in descending order of salary. |
| 43 | Demonstrate how to restrict and sort data in SQL using WHERE, ORDER BY, and DISTINCT clauses with appropriate examples.                                                                                                         |
| 44 | Demonstrate the use of single-row functions to customize query output                                                                                                                                                           |
| 45 | Illustrate how aggregate functions like SUM(), AVG(), MAX(), MIN(), and COUNT() are used along with the GROUP BY and HAVING clauses.                                                                                            |
| 46 | Discuss SQL joins for displaying data from multiple tables with an example                                                                                                                                                      |
| 47 | Analyze the purpose of SET operators (UNION, INTERSECT, and MINUS) in SQL and use examples to compare their operations and outcomes.                                                                                            |
| 48 | Explain Data Manipulation Language (DML) statements — INSERT, UPDATE, DELETE, and MERGE. Provide syntax and examples for each.                                                                                                  |
| 49 | a) Give the importance of Data Definition Language (DDL) statements in database management.<br>b) Discuss how to control user access using CREATE USER, GRANT, and REVOKE commands.                                             |
| 50 | What are sub-queries? Illustrate their use in solving complex data retrieval problems with suitable example.                                                                                                                    |



# MALLA REDDY UNIVERSITY

**Subject Name: Internet Of Things**

**Subject Code: MR25-2CB0048**

**Year & Semester: M. Tech-I&II CBCS POOL**

## **Unit-Wise Question Bank**

| <b>Qno</b>    | <b>Question</b>                                                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UNIT-1</b> |                                                                                                                                                                                               |
| 1             | Analyze the definition and characteristics of IoT and evaluate, with a real-world example, how these characteristics distinguish an IoT system from a traditional embedded system.            |
| 2             | Design the physical architecture of an IoT device for a smart-home use case and analyze the role of each functional block (sensing, processing, communication) with a labeled diagram.        |
| 3             | Analyze the IoT protocol stack layer by layer and justify, with an example deployment, the choice of protocol (MQTT/CoAP/HTTP) for a low-power sensor network.                                |
| 4             | Design the logical architecture of an IoT system for an environmental monitoring application and analyze how the IoT functional blocks interact with each other.                              |
| 5             | Compare the request-response, publish-subscribe, push-pull, and exclusive-pair communication models and analyze, with an example, which model best fits a real-time alerting IoT application. |
| 6             | Implement a conceptual design of an IoT communication API (REST-based) for a weather-monitoring device and analyze how it enables interoperability between device and cloud.                  |
| 7             | Analyze the architecture of a Wireless Sensor Network and illustrate, with an example topology, how WSN enables data acquisition in an IoT deployment.                                        |
| 8             | Analyze how Cloud Computing and Big Data Analytics are integrated in an IoT pipeline and illustrate with an example how raw sensor data is transformed into actionable insights.              |
| 9             | Evaluate the IoT levels (Level-1 to Level-6) framework and design a suitable IoT level template for a smart irrigation system, justifying your choice.                                        |
| 10            | Analyze any two domain-specific IoT applications (e.g., smart city, smart energy) and compare their architectural requirements with suitable examples.                                        |
| <b>UNIT-2</b> |                                                                                                                                                                                               |
| 11            | Analyze the concept of Machine-to-Machine (M2M) communication and illustrate, with an example architecture, how it differs from conventional IoT communication.                               |
| 12            | Design an SDN-based network architecture for an IoT deployment and analyze how it improves scalability and manageability compared to a traditional network.                                   |
| 13            | Analyze the working of Network Function Virtualization (NFV) with an example use case and justify how it reduces dependency on dedicated hardware in IoT networks.                            |

| Qno           | Question                                                                                                                                                                                                                   |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UNIT-1</b> |                                                                                                                                                                                                                            |
| 14            | Critically compare SDN and NFV for IoT networks and evaluate, with an example scenario, a combined SDN-NFV design for a large-scale IoT network.                                                                           |
| 15            | Analyze and classify the different types of sensors used in IoT systems, and design a sensor selection strategy for a smart agriculture monitoring system with justification.                                              |
| 16            | Evaluate the key characteristics (accuracy, sensitivity, resolution, response time) considered while selecting a sensor, and apply this evaluation to choose a suitable sensor for a temperature-critical IoT application. |
| 17            | Analyze the various sensing types and sensing considerations, and design a sensing strategy for a multi-parameter environmental monitoring IoT system.                                                                     |
| 18            | Analyze the role of actuators in closing the control loop of an IoT system and implement, with an example, an actuator-based automation scenario (e.g., smart irrigation valve).                                           |
| 19            | Compare and analyze different types of actuators (electrical, hydraulic, pneumatic, thermal) and justify the most suitable actuator type for a smart home automation application.                                          |
| 20            | Evaluate the important actuator characteristics that influence IoT system design and analyze their impact on the reliability of a real-time IoT control system.                                                            |
| <b>UNIT-3</b> |                                                                                                                                                                                                                            |
| 21            | Analyze the hardware architecture and pin layout of Arduino Uno and implement a circuit design with example code to interface a sensor for data acquisition.                                                               |
| 22            | Compare and analyze the I2C, SPI, and Parallel communication interfaces of Arduino Uno, and implement an example program demonstrating I2C-based sensor interfacing.                                                       |
| 23            | Implement and analyze an Arduino program using Digital I/O functions to control an LED based on a push-button input, explaining the working of each instruction.                                                           |
| 24            | Implement and analyze an Arduino program using Analog I/O functions to read data from an analog sensor (e.g., LDR or potentiometer) and display it on the Serial Monitor.                                                  |
| 25            | Analyze the I2C communication protocol in depth and implement an example program to interface an I2C-based sensor (e.g., temperature sensor) with Arduino Uno.                                                             |
| 26            | Analyze the SPI communication protocol and implement an example Arduino program to interface an SPI-based module (e.g., SD card or display) with justification of the wiring.                                              |
| 27            | Design, implement, and analyze an Arduino program to interface an LED and a push button such that the LED toggles state on every button press (debounce logic included).                                                   |
| 28            | Analyze the Arduino IDE workflow (compilation, verification, uploading) and implement a step-by-step demonstration of uploading and debugging a program on Arduino Uno.                                                    |
| 29            | Evaluate the role of built-in and third-party Arduino libraries in simplifying sensor interfacing, and implement an example using a library-based sensor (e.g., DHT11) program.                                            |
| 30            | Design and implement a complete Arduino-based temperature monitoring IoT system with example code and analyze its accuracy and limitations.                                                                                |
| <b>UNIT-4</b> |                                                                                                                                                                                                                            |
| 31            | Analyze the hardware architecture and GPIO layout of Raspberry Pi and compare it with Arduino Uno in terms of processing capability for IoT applications.                                                                  |
| 32            | Evaluate different operating systems available for Raspberry Pi (Raspberry Pi OS, Ubuntu, etc.) and justify the most suitable OS choice for a resource-constrained IoT edge application.                                   |

| Qno           | Question                                                                                                                                                                                 |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UNIT-1</b> |                                                                                                                                                                                          |
| 33            | Implement and analyze the step-by-step configuration process of Raspberry Pi (imaging OS, headless setup, network configuration) for first-time IoT deployment.                          |
| 34            | Analyze the GPIO pin layout of Raspberry Pi and design a wiring scheme with example code to interface a digital sensor for an IoT application.                                           |
| 35            | Design and implement a Python program to interface a sensor (e.g., DHT11 or ultrasonic sensor) with Raspberry Pi and analyze the data acquisition process.                               |
| 36            | Implement a Python program to blink an LED connected to Raspberry Pi GPIO pins using the RPi.GPIO library, and analyze the timing behavior of the program.                               |
| 37            | Design and implement a camera-based surveillance IoT system using Raspberry Pi and analyze its performance in terms of latency and storage requirements.                                 |
| 38            | Critically analyze the architectural and application-level differences between Arduino and Raspberry Pi, and justify which platform is suitable for a given IoT case study.              |
| 39            | Implement and analyze the process of remotely accessing and programming Raspberry Pi using SSH, including security considerations for IoT deployment.                                    |
| 40            | Design and implement an integration of Raspberry Pi with a cloud platform (e.g., ThingSpeak/AWS IoT) for real-time IoT data logging, and analyze the end-to-end data flow.               |
| <b>UNIT-5</b> |                                                                                                                                                                                          |
| 41            | Design and analyze the architecture of an Agricultural IoT system with a neat diagram, implementing an example scenario of soil-moisture-based automated irrigation.                     |
| 42            | Analyze the various components involved in an Agricultural IoT system and evaluate their integration in a real-time crop monitoring implementation.                                      |
| 43            | Critically evaluate the advantages and limitations of implementing IoT in agriculture, supporting your analysis with a practical implementation example.                                 |
| 44            | Design and analyze the architecture of a Vehicular IoT system along with its key components, illustrating with an example of a connected-vehicle traffic monitoring implementation.      |
| 45            | Evaluate the advantages and challenges of Vehicular IoT systems and analyze their impact on a real-world intelligent transportation system implementation.                               |
| 46            | Design and analyze the architecture of a Healthcare IoT system with a neat diagram, implementing an example of a remote patient-monitoring scenario.                                     |
| 47            | Analyze the various components of a Healthcare IoT system and evaluate their integration in a real-time patient vital-monitoring implementation.                                         |
| 48            | Critically evaluate the advantages of Healthcare IoT in improving patient care, supporting your evaluation with a practical implementation example.                                      |
| 49            | Analyze the risks and challenges (security, privacy, reliability) associated with Healthcare IoT systems and propose mitigation strategies with examples.                                |
| 50            | Critically compare and analyze Agricultural, Vehicular, and Healthcare IoT architectures, evaluating their unique components, implementation challenges, and applications with examples. |



## MALLA REDDY UNIVERSITY

### M.Tech. I Year - II Semester Examinations(R25) Question Bank

Subject Code: MR25-2CB0045

Subject Name: Large Language Models

| Q.no          | Question                                                                           |
|---------------|------------------------------------------------------------------------------------|
| <b>UNIT-1</b> |                                                                                    |
| 1             | Explain the evolution of NLP from n-grams to neural language models.               |
| 2             | Compare Word2Vec, GloVe, and FastText embeddings with advantages and limitations.  |
| 3             | Describe LSTM and GRU architectures and explain their limitations.                 |
| 4             | Explain the Transformer architecture with encoder and decoder components.          |
| 5             | Discuss the self-attention mechanism and scaled dot-product attention.             |
| 6             | Explain multi-head attention and its advantages over single-head attention.        |
| 7             | Describe positional encoding and its role in Transformers.                         |
| 8             | Explain pre-training paradigms: MLM, CLM, and Seq2Seq.                             |
| 9             | Compare tokenization techniques: BPE, WordPiece, SentencePiece, and tiktoken.      |
| 10            | Discuss the role of layer normalization and feed-forward networks in Transformers. |
| <b>UNIT-2</b> |                                                                                    |
| 11            | Explain the evolution of GPT models from GPT-2 to GPT-4 and their capabilities.    |
| 12            | Compare BERT and its variants (RoBERTa, ALBERT, DistilBERT).                       |
| 13            | Explain the T5 model and the text-to-text transfer framework.                      |
| 14            | Discuss open-source LLMs such as LLaMA, Mistral, Mixtral, Falcon, and Phi-2.       |
| 15            | Explain domain-specific LLMs with examples (BioMedLM, FinBERT, CodeLlama).         |
| 16            | Describe scaling laws and the Chinchilla approach to compute-optimal training.     |
| 17            | Explain Mixture of Experts (MoE) architecture with an example.                     |
| 18            | Discuss multimodal LLMs such as CLIP, DALL-E, and GPT-4V.                          |
| 19            | Explain model compression techniques: pruning, distillation, and quantization.     |
| 20            | Analyze the trade-offs between model size, performance, and efficiency.            |
| <b>UNIT-3</b> |                                                                                    |
| 21            | Explain supervised fine-tuning (SFT) process and datasets used.                    |
| 22            | Discuss instruction tuning and its role in improving LLM usability.                |
| 23            | Explain RLHF with reward modeling and PPO.                                         |
| 24            | Compare RLHF with Constitutional AI.                                               |
| 25            | Explain Direct Preference Optimization (DPO) and ORPO.                             |
| 26            | Discuss Parameter-Efficient Fine-Tuning (PEFT) techniques such as LoRA and QLoRA.  |
| 27            | Compare different PEFT methods: Prefix tuning, Prompt tuning, and IA3.             |
| 28            | Explain adapter-based fine-tuning strategies.                                      |
| 29            | Discuss continual learning in LLMs and catastrophic forgetting.                    |
| 30            | Explain the workflow for fine-tuning LLaMA or Mistral using Hugging Face           |

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|    | tools.                                                                          |
|    | <b>UNIT-4</b>                                                                   |
| 31 | Explain zero-shot, few-shot, and chain-of-thought prompting with examples.      |
| 32 | Discuss advanced prompting strategies such as ToT, ReAct, and self-consistency. |
| 33 | Explain the architecture of Retrieval-Augmented Generation (RAG).               |
| 34 | Describe dense retrieval and re-ranking in RAG pipelines.                       |
| 35 | Explain vector databases and their role in LLM applications.                    |
| 36 | Discuss document chunking and embedding strategies.                             |
| 37 | Explain advanced RAG techniques such as HyDE and fusion retrieval.              |
| 38 | Discuss evaluation metrics for RAG systems (faithfulness, relevance, recall).   |
| 39 | Explain how to build a production-grade RAG system using LangChain.             |
| 40 | Compare RAG-based systems with traditional LLM-based QA systems.                |
|    | <b>UNIT-5</b>                                                                   |
| 41 | Explain LLM agents and their components such as tools, memory, and planning.    |
| 42 | Discuss agent frameworks such as LangChain, AutoGen, and CrewAI.                |
| 43 | Explain common LLM application patterns like summarization and QA.              |
| 44 | Discuss real-world applications of LLMs in healthcare and finance.              |
| 45 | Explain hallucination in LLMs and its root causes.                              |
| 46 | Discuss techniques to mitigate hallucination in LLM systems.                    |
| 47 | Explain safety mechanisms such as bias detection and prompt injection defense.  |
| 48 | Compare different deployment strategies for LLMs (API vs local models).         |
| 49 | Discuss evaluation benchmarks like MMLU, HellaSwag, and HumanEval.              |
| 50 | Explain ethical issues, data privacy, and governance in LLM deployment.         |



# MALLA REDDY UNIVERSITY

(As per Telangana Private Universities Act No. 13 of 2020 &  
G.O.Ms.No. 14, Higher Education (UE) Department)

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Course : **M. Tech**  
Specialisation : **Computer Science / Cloud Computing / Big Data Analysis/ML & DL**  
Code : **MR23-2BS0131**  
Subject Name : **Quantum Computing**

| S.No   | Question                                                                                                                                                                                    |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UNIT-1 |                                                                                                                                                                                             |
| 1      | What is the need of quantum computing and discuss the types of computing                                                                                                                    |
| 2      | What is the future of quantum computing ?                                                                                                                                                   |
| 3      | Discuss the applications of quantum computing in artificial intelligence                                                                                                                    |
| 4      | How can quantum computing impact the field of drug discovery and materials design ?                                                                                                         |
| 5      | Discuss the potential role of quantum computing in accelerating machine learning algorithms.                                                                                                |
| 6      | Explain why quantum computers are needed in today's technological landscape. What specific computational problems exist that classical computers cannot efficiently solve?                  |
| 7      | Evaluate the current state of quantum computing technology and discuss the most significant challenges that must be overcome for practical, widespread implementation                       |
| 8      | How does quantum computing enhance deep learning capabilities beyond what's possible with classical computing? Describe the mechanisms and potential advantages.                            |
| 9      | Trace the historical development of quantum computing from its theoretical origins to present day. Identify at least three pivotal moments or breakthroughs and explain their significance. |
| 10     | Discuss how quantum computing could transform artificial intelligence research and application.                                                                                             |
| UNIT-2 |                                                                                                                                                                                             |
| 11     | What is complex vector space and discuss its properties.                                                                                                                                    |
| 12     | Describe the Dirac notation (Bra-Ket formalism) and its advantages in representing quantum states                                                                                           |
| 13     | Define Hermitian and unitary operators, and discuss their significance in quantum mechanics.                                                                                                |
| 14     | What are the Pauli operators, and how are they used in quantum computations and quantum gates?                                                                                              |
| 15     | Enumerate and explain the postulates of quantum mechanics as they apply to quantum computing.                                                                                               |
| 16     | What is a Hilbert space and how does it relate to quantum states?                                                                                                                           |
| 17     | Describe the physical meaning of the inner product between two quantum states.                                                                                                              |
| 18     | What is a tensor product and why is it used when dealing with multiple qubits?                                                                                                              |

|               |                                                                                                                                                                                    |
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| 19            | Explain what eigenvalues and eigenvectors represent in the context of quantum measurements.                                                                                        |
| 20            | List the four Pauli operators and explain what each one does to a qubit.                                                                                                           |
| <b>UNIT-3</b> |                                                                                                                                                                                    |
| 21            | (a) What is a Quantum Bit (qubit), and how is it different from a classical bit ?<br>(b) Explain the concept of superposition in quantum computing and its significance for qubits |
| 22            | Explain the concept of superposition and entanglement in quantum computing and its significance for qubits                                                                         |
| 23            | What are quantum gates, and how do they operate on qubits to perform computations?                                                                                                 |
| 24            | How does the Bloch sphere help us visualize a single qubit state? What do the different points on the sphere represent?                                                            |
| 25            | Write the general form of a single qubit in Dirac notation, showing how probability amplitudes relate to the basis states $ 0\rangle$ and $ 1\rangle$ .                            |
| 26            | Describe the action of the Pauli X gate on a qubit. Why is it sometimes called the "quantum NOT gate"?                                                                             |
| 27            | Describe the Pauli-X, Y and Z gates and its role in quantum computing.                                                                                                             |
| 28            | Explain the Hadamard gate and its ability to create superpositions in quantum circuits                                                                                             |
| 29            | Draw the circuit symbol for the Toffoli (CCNOT) gate and explain its function. Why is this gate important for quantum computing?                                                   |
| 30            | How does the SWAP gate work? Write its matrix representation and explain what happens when it's applied to two qubits.                                                             |
| <b>UNIT-4</b> |                                                                                                                                                                                    |
| 31            | Explain the concept of quantum parallelism and how it is utilized in Deutsch's algorithm.                                                                                          |
| 32            | Explain the problem that Deutsch's Algorithm solves and how it demonstrates quantum advantage. What is the classical complexity versus the quantum complexity ?                    |
| 33            | Describe the circuit implementation of Deutsch's Algorithm. What is the role of the Hadamard gates in this algorithm?                                                              |
| 34            | Describe the circuit implementation of Deutsch's Algorithm. What is the role of the Hadamard gates in this algorithm?                                                              |
| 35            | What is the main challenge in classical factoring algorithms, and how does Shor's algorithm overcome it?                                                                           |
| 36            | Explain how Grover's search algorithm achieves the quadratic speedup over classical search algorithms.                                                                             |
| 37            | Outline the key steps in Shor's Factoring Algorithm, specifically explaining the role of the quantum Fourier transform.                                                            |
| 38            | Describe the main sources of noise and errors in quantum computers. How do these differ from errors in classical computing?                                                        |
| 39            | What is quantum annealing and how it relates to finding the ground state of a                                                                                                      |

|    |                                                                                                                                                                          |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | system.                                                                                                                                                                  |
| 40 | Explain quantum noise in quantum computing. How can quantum error correction codes help preserve the integrity of quantum information in the presence of noise ?         |
|    | <b>UNIT-5</b>                                                                                                                                                            |
| 41 | Describe what Qiskit is and explain its main components. What role does each component play in quantum programming?                                                      |
| 42 | Write the Qiskit code to create a simple quantum circuit that places a single qubit in an equal superposition state and then measures it.                                |
| 43 | Explain how to visualize a quantum circuit in Qiskit. What methods are available to display both the circuit diagram and the simulated results?                          |
| 44 | What are quantum circuits, explain implementation of Pauli X gate using Qiskit ?                                                                                         |
| 45 | What are quantum circuits, explain implementation of Pauli Z gate using Qiskit ?                                                                                         |
| 46 | What are quantum circuits, explain implementation of CNOT gate using Qiskit ?                                                                                            |
| 47 | Explain implementation of Toffoli Gate using Qiskit                                                                                                                      |
| 48 | What quantum gates are involved in creating a quantum circuit for the OR gate in Qiskit                                                                                  |
| 49 | Demonstrate with Qiskit code how to implement the Hadamard gate on a qubit initially in state $ 0\rangle$ , and explain what the expected measurement outcomes would be. |
| 50 | Outline the steps to set up an IBM Quantum Experience account and explain how to access real quantum hardware through this platform.                                     |